DEVESH BHASKARAN

+91-93537-41919 | deveshbhaskaran@gmail.com

Devesh B | devesh-b | Devesh B

Bengaluru, Karnataka, India - 560033

Objective

To pursue doctoral research in advanced SoC and mixed-signal circuit design, focusing on energy-efficient architectures, low-jitter timing systems, and resilient digital interfaces. My goal is to contribute to next-generation semiconductor technologies that integrate digital, analog, and RF design principles for high-performance embedded systems.

Experience

- **Analog Devices India Pvt. Ltd.** — *June 2025 - Present*
  Digital Design Engineer | Full-Time — Bengaluru, India
  - Developed advanced 10BASE-T1S PHY, PLL, GPT, UART and enhanced PWM architectures for next-generation battery management ICs in electric vehicles.
  - Conducted analysis on system-level timing, jitter, identifying key optimizations for ISO functional safety.
- **Analog Devices India Pvt. Ltd.** — *Jan 2025 - June 2025*
  Digital Design Intern | Full-Time — Bengaluru, India
  - Developed split configuration architecture for a dual-core lockstep processor in an SoC, achieving improved fault tolerance and system reliability in safety-critical applications.
  - Implemented a functionally safe SPI module with configurable clock polarity and phase, enhancing peripheral communication flexibility and ensuring reliable data exchange across multiple device interfaces.
- **PES University** — *Jan 2025 - Mar 2025*
  Teaching Assistant – RF Microelectronics | Part-Time — Bengaluru, India
  - Designed and delivered an advanced RF microelectronics module on oscillator circuit design, developing Spectre RF simulation workflows that enhanced analysis accuracy and reduced debugging time.
  - Conducted structured Q&A and mentoring sessions that improved student engagement, concept retention, and overall project performance.
- **PES University** — *Aug 2024 - Dec 2025*
  Teaching Assistant – Computer Organization and Design | Full-Time — Bengaluru, India
  - Instructed students in RISC-V Assembly programming and guided them in designing a single-cycle RISC-V processor, conducting hands-on project evaluations that enhanced hardware design comprehension and course performance.
  - Mentored student teams to extend processor designs toward system-level performance analysis and SoC integration, fostering practical understanding of end-to-end digital system design.
- **Analog Devices India Pvt. Ltd.** — *June 2024 - July 2024*
  Digital Design Intern | Full-Time — Bengaluru, India
  - Optimized a PWM IP core by enhancing pulse position control and integrating adjustable duty cycles, improving modulation accuracy by 10%.
  - Integrated and analyzed a CAN FD architecture, increasing data rates up to 5 Mbps and enhancing communication reliability in automotive systems.

Education

- **PES University** — *Sep 2021 - Aug 2025*
  BTech. in Electronics and Communication Engineering — Bengaluru, India
  - GPA: 9.32/10.00
  - Awarded a Silver medal for being ranked 9th in the University.
- **Christ Junior College** — *Jun 2021*
  Pre-University Education — Bengaluru, India
  - Grade: 96.5%
- **Clarence High School** — *Mar 2019*
  Secondary Education — Bengaluru, India
  - Grade: 95.6%
  - Awarded Principal's Trophy for Excellence in the ICSE board exam
  - Received the Smt. Narayanikutty Menon Memorial Trophy for a perfect score in ICSE Mathematics

PROJECTS

- **Embedded SoC Design (RV32IM): Designed a RISC-V SoC with UART, SPI, and QSPI interfaces** *September 2025*
  Tools: SystemVerilog, C, Cadence Design Suite
  - Developed a RISC-V (RV32IM) core for FPGA validation with UART, SPI, and QSPI interfaces, enabling robust peripheral communication and SoC integration using SkyWater SKY130A PDK.
  - Applied digital design verification methodologies to ensure functional correctness and evaluate system-level performance.
- **8-bit RISC-V CPU: Optimized CPU for IoT applications, preparing for tapeout** *January 2025*
  Tools: SystemVerilog, C, Assembly
  - Developed an 8-bit RISC-V CPU for low-power IoT applications using energy-efficient design techniques, achieving scalable and tapeout-ready hardware for Tiny Tapeout 9.
  - Performed detailed performance analysis to optimize instruction execution efficiency and minimize power consumption.
- **6T SRAM Array Implementation: Designed and integrated a stable SRAM cell** *May 2024*
  Tools: Cadence-Virtuoso
  - Designed and implemented a 6T SRAM cell and array using the SkyWater SKY130A PDK, achieving a fabrication-ready layout with a 5% improvement in read/write stability margins.
  - Conducted circuit-level simulations to evaluate stability and performance across process, voltage, and temperature variations.
- **Layered Testbench for 4-Bit Shift Register: Structured verification environment for shift register** *January 2024*
  Tools: SystemVerilog
  - Developed a layered SystemVerilog testbench comprising generator, driver, monitor, interface, and scoreboard modules for a 4-bit shift register.
  - Applied functional verification methodologies to evaluate performance and identify design issues efficiently.
- **Design of 4-Bit Flash ADC Architecture: Designed using Op-Amp and Wallace Tree Encoder** *October 2023*
  Tools: Cadence-Virtuoso
  - Designed a 4-bit Flash ADC architecture using a 2-stage Op-Amp comparator and a Bandgap Reference Circuit for high-speed and stable analog-to-digital conversion.
  - Integrated a Wallace Tree Adder as the digital encoder to enhance overall conversion speed and performance.
- **Flexible CLA Adder Design: Modular adder for timing and area optimization** *January 2023*
  Tools: SystemVerilog
  - Developed modular 4-bit CLA blocks and multiple adder designs, optimizing timing, area, and scalability
  - Conducted circuit-level simulations and benchmarked against alternative adder architectures
- **ATM Management System: ATM simulator with facial recognition** *December 2021*
  Tools: Python
  - Developed a Python ATM simulator with GUI (Tkinter/OpenCV), MySQL integration, and facial recognition for secure transactions.

PATENTS AND PUBLICATIONS

C=CONFERENCE, P=PRESENTED, O=ONGOING

[O.1] Devesh Bhaskaran, et al. (2026). **Optimized Hardware Architecture for YOLOv8 Using Algorithmic Compression and Posit-Based Approximate Computing**. FCCM RCC Competition Submission.

[C.1] Devesh Bhaskaran, et al. (2025). **Design of a Low Phase Noise Quadrature DCO Using Dual Superharmonic Injection in 55nm CMOS for Ka-Band Applications**. *Asia Pacific Conference on Circuits and Systems(APCCAS) 2025. DOI: 10.1109/APCCAS67402.2025.11376789.*

[C.2] Devesh Bhaskaran, et al. (2025). **Performance Analysis of LC Tank VCO-Based ADC and Current Starved VCO-Based ADC**. *IEEE North Karnataka Subsection Flagship International Conference (NKCon) 2025. DOI: 10.1109/NKCon66957.2025.11345792.*

[C.3] Devesh Bhaskaran, et al. (2023). **Exploring FMU-NET: A Comprehensive Study on Person Identification through Novel Semantic Segmentation Techniques**. In *2023 International Conference on Evolutionary Algorithms and Soft Computing Techniques (EASCT)*, IEEE . 20-21 October 2023, Bengaluru, India. DOI: 10.1109 / EASCT 59475.2023.10392590

SKILLS

- **Programming Languages:** System Verilog, Verilog, C, C++, Python, Java, Embedded C, SQL, Assembly
- **Specialized Area:** Digital Circuit Design, RF Circuit Design, Analog Circuit Design, Mixed-Signal Systems.
- **Other Tools & Technologies:** Cadence Virtuoso/SpectreRF/Genus, Vivado Design Suite, QUCS, Quartus Prime, LT-Spice, CST Studio, GNS3, Wireshark, RIPES.
- **Research Skills:** Circuit Design & Simulation, Performance Analysis, Signal Processing, Embedded Systems.
- **Mathematical & Statistical Tools:** MATLAB, MATLAB Simulink, Excel, Python (SciPy, Statsmodels).

Honors, Awards and Recognitions

- **Silver Medal — 9th Rank in University** *October 2025*
  PES University
  - Awarded Silver Medal for securing 9th rank in the University during B.Tech in Electronics and Communication Engineering.
- **Best Outgoing Student — Department of ECE** *August 2025*
  PES University
  - Recognized as the Best Outgoing Student for outstanding academic performance, leadership, and contributions to the Department of ECE.
- **Tiny Tapeout Award — Chipalooza** *March 2024*
  Efabless Corporation
  - Received a $300 award and earned a slot in the Tiny Tapeout competition focused on Analog and Mixed-Signal IC design.
- **MRD Scholarship** *2022 - 2025*
  PES University
  - Earned MRD Scholarship for ranking in the top 5% of the university across Semesters III–VI.
- **Second Place — Mahil AI Hackathon** *April 2023*
  PES University
  - Developed a compact Raspberry Pi–based badge/keychain capable of detecting distress calls and transmitting the user's location to nearby phones and police stations for rapid assistance.
- **C.N. Rao Scholarship** *2021 - 2022*
  PES University
  - Earned C.N. Rao Scholarship for ranking in the top 10% of the university across Semesters I–II.
- **Third Place — HackLoop Hackathon** *October 2021*
  PES University
  - Secured 3rd place for developing an application and prototype device to assist dementia patients in improving cognitive activity and ensuring safety.
  - The device alerts relatives if the patient moves beyond a defined radius and provides real-time tracking and navigation support.

Leadership Experience

- **Workshop Instructor – SoC Architecture & Design Theory for IoT** *June 2025*
  PES University
  - Served as a Workshop Instructor for the 5-day program titled *"SoC Architecture & Design Theory for IoT"*, conducting sessions organized by the IoT MakerSpace and the Department of ECE, PES University.
  - Delivered hands-on instruction throughout the program and guided participants through core SoC architecture and design concepts.
- **Co-founder and Treasurer, IEEE RAS and Sensors Council Student Branch** *May 2024 - Nov 2022*
  PES University
  - Co-founded the student branch and managed financial planning, budgeting, and resource allocation.
  - Served as Event Head for national-level hackathon *UNIMATE* and ideathon *HealthSpark*, ensuring smooth execution.
- **President, IEEE SSCS and Photonics Student Branch** *April 2024 - April 2023*
  PES University
  - Served as Event Head for IEEE Day, hosting Prof. Shanthi Pavan (IIT Madras), and organized two technical talks and a layout hackathon in collaboration with Marvell Technology.
  - Mentored student members and guided project initiatives, promoting hands-on learning and innovation.
- **Mentor Head, InFeynite Hackathon** *September 2024 - June 2024*
  Department of Electronics and Communication, PES University
  - Led the mentorship team and guided participants throughout the hackathon.
  - Managed technical sessions and supported participants in developing their projects.

Professional Memberships

- **IEEE (CAS, SSCS, MTTS) Member,** Membership ID: 98147149 *August 2022 - Present*

CERTIFICATIONS

- **IEEE Bangalore Section -** IEEE Webinar on Pre-Silicon RTL Verification using UVM *March 2025*
- **Department of ECE, BIT Mesra -** Webinar on Science Narrative Twisted Radio Waves *March 2025*
- **ABV-IIITM Gwalior -** Analog IC Design: Schematic to Layout Workshop *December 2024*
- **Department of Telecommunications (DOT), India -** BHARAT 6G VISION *September 2024*
- **CeNSE, IISc -** CeNSE Summer School on Semiconductor Technology and Microfabrication *June 2024*
- **CHIPS, PES University -** Standard Cell Design, Characterization And Synthesis *January 2024*
- **QuaNaD, PES University -** Semiconductor Device Modelling And Simulation *December 2023*
- **NPTEL -** VLSI Design Flow: RTL to GDS *October 2023*

ADDITIONAL INFORMATION

Languages: English(Advanced), Hindi(Intermediate), Telugu(Intermediate), Kannada(Intermediate), Tamil(Novice)

REFERENCES

1. **Dr. Shashidhar Tantry**
   Professor, Department of Electronics and Communication Engineering
   PES University, Bengaluru, India
   Email: stantry@pes.edu
   Relationship: Research Advisor, Principal Investigator - IoT MakerSpace

2. **Anjana Priya Sistla**
   Principal Architect, Automotive Battery Management Systems
   Analog Devices India Pvt. Ltd., Bengaluru, India
   Email: anjanapriya.sistla@analog.com
   Relationship: Research Advisor and Mentor

3. **Prof. Dhanashree G Bhate**
   Assistant Professor
   PES University, Bengaluru, India
   Email: dhanashreegb@pes.edu
   Relationship: Academic and Research Mentor